

proximal-policy opt
 RL algorithms: PPO $\rightarrow$ from reinforcement learning
 direct-preference opt
 GRPO: remove value functions
 group-relative policy optimization

Overview of course logistics: Efficiency drive design
 how to sequence the most compute from a given hardware decision.

- #1: Data preprocessing: avoid waste time on data cleaning
- #2: Tokenization: working with raw bytes is promising but current model architecture doesn't fit in.
- #3: Model Architecture: model variants to reduce memory or steps
- #4: Training / model efficiency: get away from one single epoch
- #5: Scaling laws / interenergy: less compute on smaller model
- #6: Alignment: human preference desired for fine-tune / post-training

summary
 get away from one single epoch
 tune model $\rightarrow$ If tune model to required case, use smaller models

design decision $\Leftarrow$ AI system

changing

Tomorrow: data-constrained

1 Tokenization: Different ~~and~~ LLMs have different tokenizer approaches

~~no~~

~~space~~ how many spaces are considered as characters a token
~~number~~ how many numeric letters are considered as a token.

Compression ratio: 1 token = ~~1.6~~ byte
1.6

Byte-tokenizer ~~character~~-tokenizer

① word-tokenizer

~~character~~ Byte-pair encode - tokenizer



GPT 2, 3 ... etc --

Start with each byte as a token,
and then merge the most common
pair of adjacent tokens